

FROM ORDERS TO DECISIONS

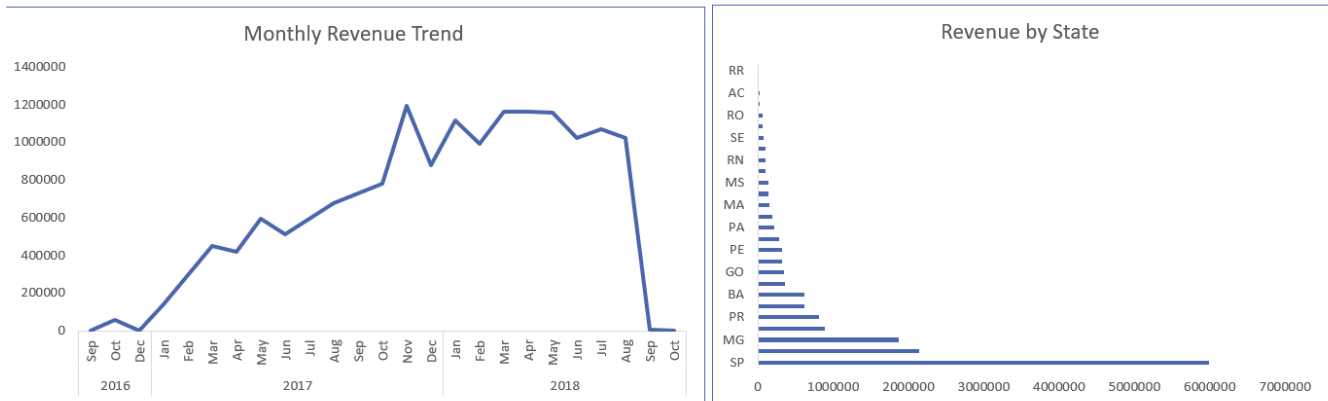
Olist E-Commerce Business Analysis | Business case study

A practical analysis of commercial performance, customer retention, delivery reliability and customer experience using the Olist Brazilian e-commerce dataset.

OLIST E-COMMERCE EXECUTIVE DASHBOARD

Business Performance Overview

TOTAL REVENUE	TOTAL ORDERS	AVERAGE ORDER VALUE	REPEAT CUSTOMER %
₹16.01M	99441	₹160.99	3.12%



THE BUSINESS QUESTION

If I were reviewing this marketplace as a business analyst, I would not stop at total sales. I would want to know where revenue is coming from, whether customers come back, whether delivery problems are affecting experience, and which parts of the marketplace deserve attention.

OBJECTIVE

Build a connected view of the marketplace and turn the transaction data into a small set of decisions that a commercial or operations team could act on.

WHAT I WORKED WITH

ORDERS	CUSTOMERS	ORDER ITEMS	SELLERS	REVIEWS
99,441	96,096	112,650	3,095	98,410

The analysis uses seven core Olist tables covering orders, customers, order items, products, sellers, payments and reviews. The data spans 2016–2018.

HOW I APPROACHED IT

Step	What I did
1. Frame	Defined questions around revenue, retention, delivery, products, sellers and satisfaction.
2. Model	Connected the seven tables using customer, order, product and seller relationships.
3. Analyze	Used PostgreSQL joins, aggregations, CASE logic, CTEs and window functions.
4. Validate	Cross-checked KPI outputs and avoided treating incomplete end-period data as a genuine business drop

5. Communicate	Built six Excel dashboards so each business area can be reviewed separately.
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The result is not a single dashboard. It is a connected analysis: the same order can be traced from customer and seller to product, payment, delivery outcome and review.

WHAT THE DATA ACTUALLY SAYS

Four findings that changed the way I would look at the business

1 | A SMALL REPEAT-CUSTOMER BASE IS GENERATING DISPROPORTIONATE VALUE

UNIQUE CUSTOMERS	REPEAT CUSTOMERS	REPEAT RATE	REPEAT-CUSTOMER REVENUE
96,096	2,994	3.12%	8.78%

Only 3.12% of customers are repeat customers, yet this group contributes 8.78% of revenue. That gap makes retention worth investigating before simply trying to acquire more customers.

Next analysis: cohort retention, time to second purchase, customer value by first-order category and acquisition-to-repeat conversion.

2 | DELIVERY IS THE CLEAREST EXPERIENCE SIGNAL IN THE DATA

ON-TIME DELIVERY	DELAYED DELIVERY	AVG. DAYS LATE	DELAYED REVENUE
91.88%	8.11%	9.55	8.44%

The rating difference is large: on-time orders average 4.30/5, while delayed orders average 2.57/5. That is roughly a 40% lower rating for delayed orders.

This is an association, not proof of a single root cause. The useful business question is therefore: where are delays concentrated, and what operational factors explain them?

3 | SCALE AND CUSTOMER SATISFACTION DO NOT ALWAYS POINT TO THE SAME PLACE

Measure	Finding
Highest revenue category	Beauty & Health
Highest units sold	Bed Bath Table
Overall average rating	4.09 / 5
Lowest category average rating	Office Furniture — 3.49 / 5

Looking at only revenue would hide this difference. Category decisions should be made using revenue, volume, revenue per unit and customer rating together.

4 | SELLER PERFORMANCE NEEDS MORE THAN A DELAY RANKING

I compared sellers on revenue, revenue per unit, average rating and delivery delay. The analysis showed that delivery delay alone did not fully explain differences in seller ratings. Category mix showed a stronger association with rating differences in the analyzed data.

That changes the next question from “Which sellers are late?” to “Which seller-category combinations are creating the biggest commercial or experience risk?”

ONE IMPORTANT DATA-CHECK

The monthly revenue series has an incomplete tail in September–October 2018. I therefore avoid presenting that sharp drop as a genuine collapse in business performance. This is a validation step I would perform before taking a dashboard number into a business meeting.

FROM FINDING TO ACTION

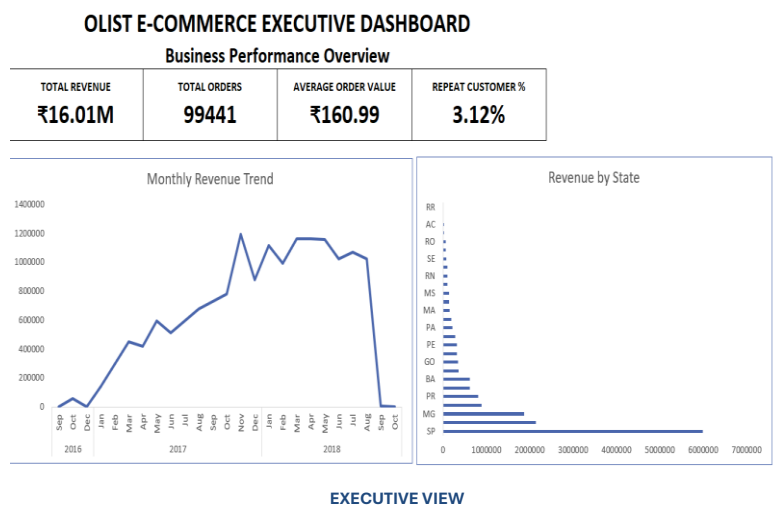
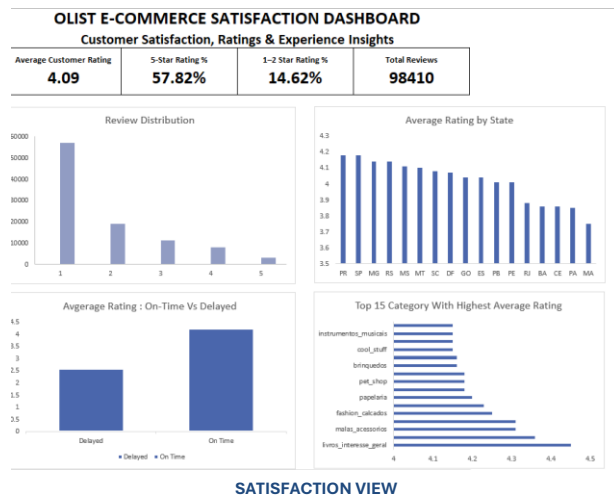
What I would investigate next — and how the dashboard supports it

RECOMMENDED BUSINESS FOLLOW-UPS

Area	What the analysis suggests	Next investigation
Customer	Repeat customers are small in number but high in value.	Cohorts, second-purchase timing, customer segments.
Operations	Delays coincide with substantially lower ratings.	Delay hotspots by seller, state, category and period.
Products	Revenue leaders and volume leaders differ; ratings vary by category.	Product-level reviews, category mix and revenue/volume trade-offs.
Sellers	Delay does not fully explain rating differences.	Seller × category performance and review-level diagnostics.

THE REPORTING LAYER





TOOLS & DELIVERABLE

PostgreSQL → relational analysis and KPI computation | Excel Data Model | PivotTables/Pivot Charts → dashboards | Six stakeholder views → Executive, Customer, Operations, Product, Seller and Satisfaction.

PROJECT TAKEAWAY

The strongest outcome is not a single number. It is the ability to connect commercial, customer and operational signals and turn them into a clear next question. In this dataset, retention and delivery experience stand out as the two areas I would prioritize for deeper analysis.

Project repository: github.com/Ayushvj14/olist-ecommerce-business-analysis

Built as an individual portfolio project